

Improving Transparency In Machine Learning Based Intrusion Detection Systems Using Explainable AI

1st Parth Tomar

Department of Computer Science Engineering
Bennett University
Greater Noida, India
parthtomar1910@gmail.com

2nd Pranjal Srivastava

Department of Computer Science Engineering
Bennett University
Greater Noida, India
pranjalsrivastava.2024@gmail.com

3rd Arun Agrawal

Department of Computer Science Engineering
Bennett University
Greater Noida, India
arunpragrawal@gmail.com

I. INTRODUCTION

As the usage of the Internet and communication through digital channels have increased, the rate at which cyber attacks are increasing is a significant issue for organizations, institutions, and network administrators. Intrusion Detection Systems (IDS) are popular tools that can be utilized to trace network traffic and detect activities as malicious or not. Machine learning and deep learning technologies have made considerable progress in recent years, which allowed enhancing the performance of IDS by allowing the automatic identification of complex attack patterns within the network traffic. Such models process huge amounts of network traffic and determine traffic as normal or malicious using previously learned patterns. It has been established in a number of studies that machine learning algorithms such as random forest, decision trees, and neural networks have the capability of providing high detection accuracy when it comes to detecting cyber attacks. While working on this research paper, an IDS based on Random Forest is used along with the NSL-KDD dataset. This was done to measure its effectiveness in identifying different types of cyber attacks. Although such systems are effective, a significant number of machine learning-based intrusion detection systems are black-box models according to the definition, that is, the internal decision-making process of such systems is hard to understand. Although these models can be used to successfully identify attacks, they frequently cannot be used to identify why a certain network activity has been deemed malicious. This absence of transparency is a significant problem for cybersecurity experts as they should be able to confirm these alerts by interpreting the root source and intention of these activities. This lack of reasoning in machine learning models presently in usage, as elaborated by recent studies, may decrease the trust in the automated

security system and limit its use to the real-life setting. This is a common scenario of a student studying in technical schools. Tasks related to networking are frequently carried out by students such as downloading datasets, simulation of networks, or the experiments in cybersecurity lab practicals. These activities often lead to creation of traffic patterns which are suspicious to intrusion detection systems. Lacking any explanations of the decisions made by the model, the administrators might struggle to identify whether the identified activity is a real cyber attack or a valid academic activity. As such, the security monitoring requires enhancement of the transparency of the intrusion detection systems. To overcome this problem, scientists have paid more attention to the Explainable Artificial Intelligence (XAI) methods that are used to interpret the machine learning models. In this research, SHAP is used to understand the influence of varying features on the model's decision, providing both overall and instance level explanations. SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model Agnostic Explanations) are methods that can give explanations based on the individual features that explain the model predictions. The techniques enable security analysts to know what the network characteristics add towards identifying malicious activities. Many recent research papers have demonstrated that the usage of explainable AI in combination with intrusion detection systems leads to improved transparency, increased reliability, and trustworthiness of such automated cybersecurity systems considerably. However, less frequent attack types like Remote-to-Local (R2L) and User-to-Root (U2R), still require further attention as they affect detection performance, leading to challenges categorized as class imbalance. The approach proposed in this paper is expected to work on both aspects which include accurate detection as well as improved transparency. Thus, this performance gap of interpretability can be reduced by applying

the concept of XAI to ML-based IDS.

II. LITERATURE REVIEW

A. *Intrusion Detection Systems and the Need for Explainable Artificial Intelligence*

Intrusion Detection Systems (IDS) are part of the contemporary cybersecurity systems, as they are used to monitor traffic in a network and identify activities as malicious or normal. Conventionally, the IDS systems were based on signature-based or rule-based detection whereby the known attack patterns were stored as a database. Although this method is effective in the presence of the threats that have been previously identified, it does not usually work when new or unknown attacks are involved. With increased sophistication in cyber attacks, researchers began incorporating machine learning methods in IDS systems, to improve the models capability of identifying complex attack patterns. Network traffic in a university or other learning institutions can be very diverse, with thousands of students using the same network for their needs. Students are involved in activities such as downloading data, using software tools to complete assignments, or using cloud based services to complete projects. Part of these may not be identified as normal by conventional security systems. Due to such dynamic behavior, machine learning-driven IDS models have turned out to be a desirable option to identify suspicious activities in a more intelligent method. Nevertheless, even though machine learning models provide a higher quality of detection, most of such systems are black boxes. They are able to categorize network traffic as malicious or normal the only thing that they do not explain is how they make such decisions. Such transparency may present problems to the security analysts who should know the reason why a system flagged any network activity.

B. *Emergence of Explainable Artificial Intelligence*

The issue of interpretability of the models prompted the researchers to study the field of Explainable Artificial Intelligence (XAI) Bovenzi et al. [1] carried out one of the first research works that demonstrated the relevance of explainability in artificial intelligence systems. This article provides us a detailed and justified summary on explainable AI methods and highlights the need to incorporate transparency through these models in machine learning models. It was argued by the authors that although certain machine learning models such as random forest and support vector machines can be very predictive, they do not have an internal decision-making process that humans can easily decipher. This constraint is especially crucial in cybersecurity-related tasks when the analyst has to be aware of the logic behind the alert before acting. In the absence of this interpretable behaviour, security professionals might struggle to know whether an anomaly that has been detected is an actual attack or just an unusual but valid network activity. As an example, during laboratory work, students specially in cyber security often scan networks and analyze traffic using nmap, nikto etc. as part of their academic work. These activities could produce abnormal patterns of

traffic. When a machine learning model flags this type of traffic as suspicious without providing any justification on why it was identified that way, administrators may not be able to know whether the alert is due to a real threat or a mere academic experiment.

C. *SHAP and the Development of Interpretable Machine Learning*

The SHAP (SHapley Additive exPlanations) framework provided by Patel et al. [2] can be considered one of the most impactful works to explainable AI. SHAP relies on the ideas of the cooperative game theory and estimates the value of the contribution made by each feature to the prediction of a model. SHAP gives information on how various inputs are related to the final decision made by machine learning model. This is done by assigning values of contributions to features. With the introduction of SHAP, researchers have been able to gain a lot of insight into the complexity and justifiability of machine learning models. Rather than merely coming out with a forecast, models that make use of SHAP explanations are in a position to point out which attributes played the greatest roles in determining suspicious network activity. To give an example, SHAP can be used in intrusion detection dataset to show that aspects like packet rate, duration of connection or protocol type were significant in determining traffic as malicious. Cybersecurity analysts are one group of people who find this form of interpretability helpful since it allows them to comprehend how automated security alerts arrived at the conclusion they did.

D. *Explainable AI in Intrusion Detection Systems*

Following the release of SHAP and other explainable artificial intelligence methods, scientists have started implementing explainable AI to intrusion detection systems. Indicatively, Ahmed et al. [3] performed a survey on explainable intrusion detection systems and examined the outcome of application of explainability techniques such as SHAP to machine learning based intrusion detection systems. Their experiment revealed that most of the IDS models have high detection rates, but due to the inability to interpret the results, their usefulness may be restricted by this factor. In many cases, security analysts must be in a position to tell why a certain alert was given in order to decide whether the alert is a legitimate attack or a false positive. Explainable AI methods can be used to overcome this problem by establishing the most important features and relating them to their contribution towards the decision. This is because these findings prompted researchers to develop IDS models that integrate machine learning algorithms with explainability methods. These systems are not only intended to identify cyber attacks but make meaningful explanations about the conclusions that would help the analyst to comprehend how the model arrived at its conclusions.

E. *Explainable Intrusion Detection in IoT Networks*

As the Internet of Things (IoT) devices are rapidly growing, the focus of intrusion detection research has also turned into

the protection of the IoT environment. IoT networks produce voluminous data and in many cases, the devices that make the networks have limited security measures and are susceptible to cyber attacks. Alshamrani and Alhaidari [4] have suggested an explainable intrusion detection framework dedicated to the IoT setting in order to overcome these challenges. Their method involved SHAP-based interpretations on how feature importance is utilized by machine learning models to detect attacks to the IoT network. The authors showed that explainable AI methods might be used to determine the most prominent network features that reveal malicious traffic in the IoT. Nevertheless, they have also pointed out issues like complexity in computation in implementing explainability approaches in large scale IoT networks.

F. Deep Learning and Explainable IDS Models

Besides conventional machine learning, scholars/researchers have studied how deep learning models can be applied in intrusion detection. Recurrent neural networks such as Long Short-Term Memory (LSTM) networks are especially useful in the analysis of the sequential data such as network traffic pattern. As a case in point, Lee et al. [5] suggested an attention-driven LSTM model on intrusion-detection assignments. Although the model was highly successful in detecting, the authors admitted that the deep learning models can be even harder to interpret in comparison to the traditional machine learning models. In order to solve this problem, they implemented SHAP explanations to demonstrate that how does a particular feature contributes towards the decision. Their findings suggested that combining deep learning models with explainable AI methods may improve detection performance as well as enhance the model intelligibility.

G. Comparison of Explainability Techniques

The other significant field of study is the comparison of various explainability methods. LiME (Local Interpretable Model-Agnostic Explanations) by Gupta and Sharma [6] describes individual predictions based on approximation. This is done by combining complex models with simpler interpretable models within a small set of the input data. The IDS models based on machine learning were later compared in the literature to interpret their outcome when used in combination with LIME and SHAP. These comparisons tended to yield that although both of these techniques are beneficial, SHAP often provided more consistent explanations. So it may be a preferred option in most research papers.

H. Summary of Literature

In general, the literature that has been reviewed demonstrates that the effectiveness of intrusion detection systems has been largely enhanced by machine learning and deep learning methods. The interpretability of such models however is a big challenge. Elucidable AI frameworks like SHAP and LIME have become a promising solution to enhancing transparency in an IDS framework. These methods enable the researcher and security analysts to know the variables

that affect model predictions and hence enhance confidence in automated security systems. Although this field has been advanced, more research is still required to come up with intrusion detection frameworks that are accurate in detecting as well as providing reliable interpretability. This grounds the necessity to further investigate explainable AI methods in the field of cybersecurity. [2]–[4], [7]–[17].

III. PROPOSED METHODOLOGY

The proposed study would create a machine learning-based intrusion detection system (IDS) and use the methods of Explainable Artificial Intelligence (XAI) to enhance the predictions of the model, making them more transparent and understandable. The methodology comprises a number of steps such as data collection, preprocessing data, model training, applying explainable methods, and then result assessment.

A. Dataset Selection

A publicly available dataset of intrusion detection (e.g. NSL-KDD, CICIDS2017, or UNSW-NB15) will be utilized in the first stage. These data sets comprise labeled network traffic logs of the normal network activity and various forms of cyber attacks including Denial-of-Service (DoS), brute force attacks and infiltration attacks. The benchmark datasets would guarantee that the proposed system can be tested and compared to the used intrusion detection models.

B. Data Preprocessing

The dataset will be pre-processed using a few steps before training the machine learning model to enhance the quality of data and model performance. These steps include:

- Eradication of redundancy or conflicting records.
- Handling missing values
- Encoding of categorical features to numerical ones.
- Normalizing/Scaling numerical attributes.
- Division of the data into training and testing
- Preprocessing of the data is a necessary process to guarantee that the information is clean, balanced and that the dataset is machine learnable.

C. Feature Selection

Network traffic data is usually associated with a significant number of features, and some of them may not be essential to attack detection. Correlation analysis or feature importance ranking methods will be used for feature selection as a way of determining the most relevant features. The process of eliminating irrelevant features will probably improve the models efficiency, as well as lower the computational complexity, and the overall performance.

D. Model Training

Once the data goes through preprocessing, and relevant features are selected, machine learning models such as Random Forest and Decision Tree classifier will be trained on the ready data. Such models will be trained on the network traffic data and will label each network activity as normal or malicious.

Random Forest is especially helpful since it demonstrates a high level of classification and is able to work with extensive datasets.

E. Model Evaluation

Standard metrics that are usually employed in the intrusion detection studies will be used to evaluate the trained models. These metrics include:

- Accuracy - evaluates the general figures of predictions.
- Precision - is the fraction of correctly detected attacks.
- Recall - the model has the capacity to identify actual attacks.
- F1-score - is a balance between recall and precision.

These evaluation metrics are used to find out the effectiveness of the intrusion detection model to detect malicious network activities.

F. Explainable AI Analysis

Explainable Artificial After training and analyzing the machine learning model, Explainable Artificial. Such intelligence methods as SHAP (SHapley Additive Explanations) will be used to interpret the predictions made by the model. The SHAP values also give a numerical value of the contribution made by each feature in the final prediction of the model. With SHAP analysis one is able to:

- Determine the most powerful network traffic features.
- Know the reasoning behind why some traffic has been flagged as malicious.
- Give explanations on models in a way that is interpretable

G. Result Analysis

Lastly, the findings of the explainability analysis will be evaluated to come up with insights into the correlation between network properties and attack detection. The analysis will emphasize the most important traits towards detecting malicious activities, and demonstrate how explainable artificial intelligence can enhance the efficiency of machine learning-based intrusion detection systems in terms of transparency and reliability.

IV. IMPLEMENTATION, EXPERIMENTATION AND RESULTS ANALYSIS

A. Introduction

This chapter is a detailed description of the design implementation of the Network Intrusion Detection System (NIDS) proposed, as well as the laboratory setup, testing procedure and a thorough discussion of the findings that were reached. The system uses a Random Forest classifier trained on the NSL-KDD benchmark dataset and uses SHAP (SHapley Additive explanations) to give transparent, human interpretable explanations of all the predictions made by the model. The chapter is structured as follows: Section 4.2 presents the experimental setting and data preparation; Section 4.3 outlines feature engineering and selection; Section 4.4 outlines model training and hyperparameter optimization; Section 4.5 discusses the classification performance in terms of quantitative measures

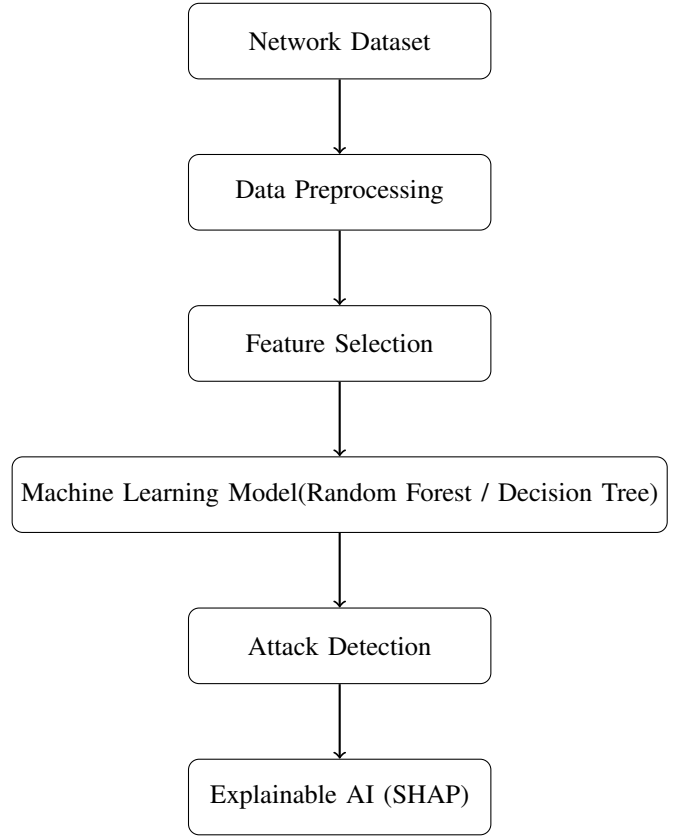


Fig. 1. Proposed Methodology Flowchart

and visualisations; Section 4.6 discusses the SHAP-based explainability results; and Section 4.7 provides the overall discussion of the findings.

B. Experimental Setup and Dataset Preparation

1) *Dataset: NSL-KDD:* All experiments were performed using the NSL-KDD dataset which is a refined version of the original KDD Cup dataset, 1999. It removes duplicate records which artificially inflated the performance of classifiers in the original corpus, and it is therefore a better benchmark in the study of intrusion detection. Each record of network connection has 43 features, including connection-level features (duration, protocol type, service, flag), content-based features (number of failed login attempts, root-shell access, hot indicators), and host-based traffic features. The official pre-partitioned training and test splits were used so that it would be reproducible and comparable to previous work. The data characteristics are summarised in Table 1.

One of the peculiarities of this dataset is a strong imbalance in classes. The two categories, R2L and U2R, comprise less than 1 % of the total training samples, but are associated with some of the most threatening real-world attacks - remote-to-local exploits and user-to-root privilege escalation. This disparity is a direct consequence on the performance of the classifiers on such minority classes as is observed in the results in Section 4.5.

Attribute	Training Set	Test Set
Total Samples	125,973	22,544
Number of Features	43	43
Normal Samples	67,343 (53.5%)	9,711 (43.1%)
DoS Samples	45,927 (36.5%)	7,458 (33.1%)
Probe Samples	11,656 (9.3%)	2,421 (10.7%)
R2L Samples	995 (0.8%)	2,885 (12.8%)
U2R Samples	52 (0.04%)	67 (0.3%)

TABLE I
NSL-KDD DATASET SPLIT AND CLASS DISTRIBUTION

2) *Label Consolidation*: NSL-KDD training set consists of 23 different attack sub-types. These were further grouped into five semantically meaningful super-categories, namely: Normal, DoS (Denial of Service), Probe, R2L (Remote-to-Local) and U2R (User-to-Root), and an Other category was added to absorb remaining classifications. This mapping is reflective of the accepted taxonomy employed in the intrusion detection literature and enables the model to learn generalizable attack signatures as opposed to overfitting to localized, limited, sub-types.

C. Feature Engineering and Selection

1) *Pre-processing*: The protocol-type, service and flag categorical features were coded with LabelEncoder and the encoder was trained and tested together on the combined training and test set, to guarantee consistent integer mappings between the train and test sets. StandardScaler was then used to standardise numeric features with zero mean and unit variance so that large absolute magnitudes of features do not skew the tree-split criterion.

2) *Univariate Feature Selection*: The SelectKBest algorithm using the ANOVA F-statistic (f-classif) was run on the entire 43 dimensional feature space to jointly select the 20 most discriminative features. The f-statistic based selection is particularly applicable to multi-class problems as it is able to measure the ratio of between-class variance to within-class variance per feature separately. Table 2 shows the chosen characteristics.

#	Feature Name	#	Feature Name
1	flag	11	dst_host_count
2	hot	12	dst_host_srv_count
3	logged_in	13	dst_host_same_srv_rate
4	count	14	dst_host_diff_srv_rate
5	serror_rate	15	dst_host_same_src_port_rate
6	srv_error_rate	16	dst_host_srv_diff_host_rate
7	error_rate	17	dst_host_serror_rate
8	srv_error_rate	18	dst_host_srv_serror_rate
9	same_srv_rate	19	dst_host_srv_rerror_rate
10	diff_srv_rate	20	srv_diff_host_rate

TABLE II
TWENTY FEATURES SELECTED BY SELECTKBEST (ANOVA F-STATISTIC, $k = 20$)

The feature set chosen is dominated by the host-based and session-based traffic statistics (dst-host-* features) and the error-rate indicators (serror-rate, error-rate). This is in line with the literature: DoS attacks are typified by abnormally

large connection counts and error rates whereas Probe attacks produce abnormally diverse destination host service patterns. The flag option, that indicates the TCP connection state, proved to be consistently significant in all types of attacks, as confirmed by the SHAP analysis in Section 4.6.

D. Model Training and Configuration

Three reasons why a Random Forest classifier was chosen as the main model are: (i) it is resistant to overfitting by default through bootstrap aggregation; (ii) it supports mixed types of features and non-linear decision boundaries without the need to assume any distribution; and (iii) it is directly compatible with tree-based SHAP explainers, allowing the computation of Shapley values exactly and not approximate. Table 3 summarises the hyperparameter configuration used.

Hyperparameter	Value / Setting
Number of estimators (n_estimators)	100
Maximum tree depth (max_depth)	20
Minimum samples to split a node	5
Feature sampling strategy (max_features)	'sqrt' (default)
Class weighting	None (uniform)
Parallelism (n_jobs)	-1 (all CPU cores)
Random seed	42

TABLE III
RANDOM FOREST HYPERPARAMETER CONFIGURATION

The depth of 20 was set to give enough complexity to gain multi-feature interaction patterns typical of the attack behaviour and the minimum-samples-to-split of 5 was set to avoid overfitting of the minority classes at the leaf level. No class weighting was done at this point; rather the imbalance analysis is left to be discussed in the Section 4.7 as a guideline to future improvement.

E. Classification Performance

1) *Overall Accuracy*: The trained Random Forest classifier had a test set accuracy of 71.27% on the NSL-KDD test set (22,544 samples). Although this number is small in itself, it should be put into the context of the overwhelming inequality in the proportion of test samples to training samples: R2L is 12.8% of test samples but less than 1% of training samples, which introduces a considerable domain shift between the train and test distributions which biases the recall of the minority classes.

2) *Per-Class Performance*: Table 4 presents the full per-class precision, recall, and F1-score.

Several key observations emerge from Table 4:

- The strongest is DoS detection (F1 = 0.81) which is fuelled by high precision (0.89) in that the model has the capacity to properly record flood-based attacks where the model is committed to predicting a DoS and the recalls at 75% show the model is successful in doing so. correctly flags three quarters of actual DoS traffic.
- The normal traffic recall is extremely high (0.93), that is, the system recalls 93% of legitimate traffic, which is a

Class	Precision	Recall	F1-Score	Support
DoS	0.89	0.75	0.81	7,458
Normal	0.62	0.93	0.74	9,711
Other	0.00	0.00	0.00	2
Probe	0.84	0.59	0.69	2,421
R2L	0.86	0.01	0.02	2,885
U2R	0.60	0.04	0.08	67
Macro Avg	0.64	0.39	0.39	22,544
Weighted Avg	0.77	0.71	0.67	22,544

TABLE IV

PER-CLASS CLASSIFICATION REPORT ON THE NSL-KDD TEST SET

vital operation requirement to reduce false positive which would inconvenience normal users.

- Probe detection performances are reasonable ($F1 = 0.69$, precision = 0.84), suggesting that scanning behaviour does leave recognisable statistical markers in the chosen 20 feature set.
- R2L and U2R classes have very low recall (0.01 and 0.04 respectively) because they are under-represented terribly in training. This model was not trained to have enough discriminative patterns over these classes, and it would make predictions of Normal in the ambiguous cases.

3) *Confusion Matrix Analysis*: The confusion matrix (see figure 2) gives a visual representation of the entire classification error pattern of all six classes.

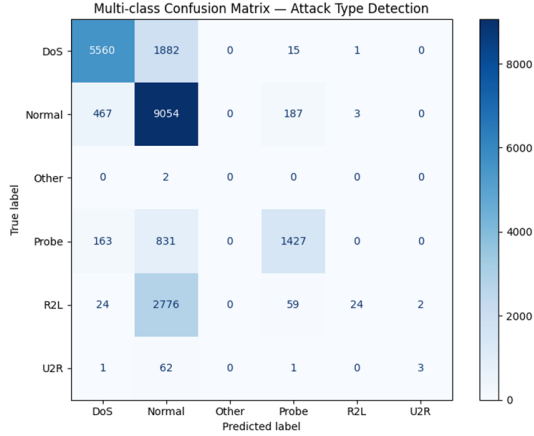


Fig. 2. Confusion matrix of the Random Forest classifier on the NSL-KDD test set. Darker diagonal cells indicate higher correct-classification counts.

As the confusion matrix shows, the most frequent error pattern is that the R2L samples (2,885 test instances) are incorrectly classified as Normal. This aligns with the imbalance of the training set - the model only trained on 995 R2L examples, which is not enough to learn the finer details of the authentication-failure patterns that can be used to distinguish between R2L and benign sessions in the feature space. This operational misclassification is severe, since it reflects false negative of intrusions based on authentication. DoS misclassifications are mainly concentrated around the Normal class (false negatives) whereas normal misclassifications are mostly inherent to the DoS category - indicating that the decision

boundary between the two classes in the model is well defined but not perfect, especially with low intensity DoS traffic that looks like legitimate burst traffic.

F. SHAP-Based Explainability Analysis

The SHAP values were calculated with the TreeExplainer on a stratified sample of 600 test instances, which produced a SHAP tensor of size (600 20 5) - (600 samples, 20 features, 5 detectable classes) - U2R was not studied because of the lack of representation in the test-set to estimate SHAP values reliably. Values SHAP are used to measure the additive contribution of each feature to the model output to a particular prediction. A positive value of SHAP between a pair of features and a class pushes the prediction towards that particular class; a negative value pushes it in the opposite direction. The size is a measure of the strength of evidence.

1) *Global Feature Importance (SHAP Summary)*: Figure 3 shows the average absolute SHAP values per class, summed up over all 600 test samples and summed up over all 20 features. This gives a worldwide perspective of what features had the greatest impact on the decisions of the classifier.

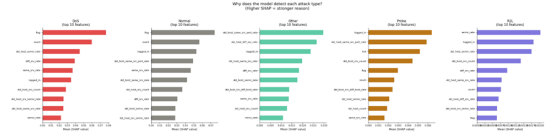


Fig. 3. SHAP summary plot showing mean —SHAP— per class. The flag and count features dominate across DoS and Normal classes.

The single most important feature on a global scale, with $SHAP = 0.077$ in DoS and 0.074 in Normal predictions, is the flag feature (TCP connection state encoding). This can be understood as follows: attack-indicative patterns of handshake failures are simply encoded in the various states of TCP connections (e.g. S0 in the case of SYN-without-response typical of SYN floods, SF in the case of normally terminated connections). The second most significant feature to detect DoS ($SHAP = 0.060$) is the count feature (number of connections to the same host in the previous two-second window), which is a direct measure of the high connection rate that is characteristic of denial-of-service behaviour.

2) *Per-Class SHAP Bar Chart*: A parallel bar chart of mean absolute SHAP values of each attack class, as shown in Figure 4, allows direct comparison of feature importance across the classes.

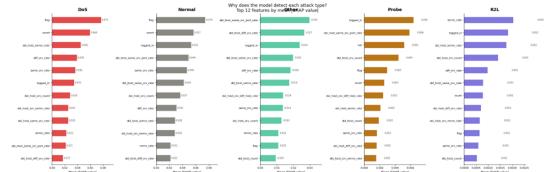


Fig. 4. SHAP bar chart showing mean —SHAP— per feature broken down by attack class. This figure is intended for direct inclusion in the thesis.

The bar chart shows the profiles of feature importance by each of the classes, which validates the fact that there are

different classes of attack types that exploit different features of the network connection:

- DoS: flag and count prevail, which are TCP state abnormalities and volumetric flooding.
- Normal: flag, count, and logged-in are equal as they represent the balanced connection image of the legitimate traffic.
- Probe: probe logged in and probe dst-host-same-src-port-rate are the most common, which are consistent with the scanning behaviour that creates diverse source-port patterns without any authenticated session being established.
- R2L: the magnitudes of SHAPs across the board are extremely low (≤ 0.003), and that is why the recall is so low (almost zero) the model has not acquired strong feature evidence on this class.

3) SHAP Beeswarm Plots (Per-Class Distribution):

Beeswarm plots are an expansion of bar-chart view that depict the complete distribution of SHAP values on a sample-by-sample basis, coloured by the standardised value of the feature. This does not only show the average importance but also the directionality and spread of the impact of each feature. DoS Class

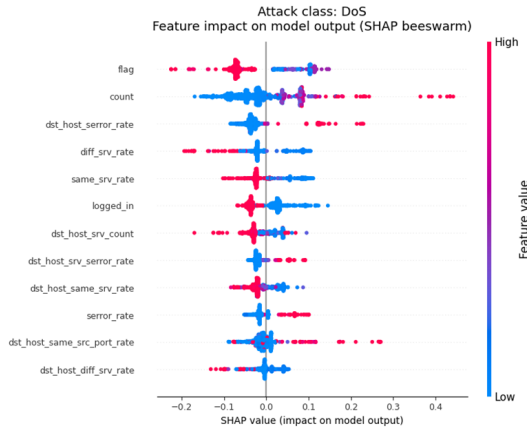


Fig. 5. SHAP beeswarm plot for the DoS class. High feature values (red) for count and dst-host-serror-rate strongly increase the DoS SHAP output.

In DoS traffic, flag and count are highly polarised beeswarm distributions, with large flag values (red, implying SYN-flood TCP states) and large count values concentrated around large positive SHAP values, with prediction forces being strongly biased towards DoS. A similar tendency is observed with the dst-host-serror-rate feature where large error rates in destination-host sessions are a characteristic attribute of DoS flooding.

Normal Traffic Class

The logged-in feature indicates a clean, high-SHAP cluster of high values (red) is a direct result of authenticated, successfully logged-in sessions being the most potent positive indicator of Normal traffic. The flag feature exhibits a bimodal distribution: the SF (normal full connection) TCP state corresponds to high positive SHAP in case of Normal and the S0 (connection attempt, no response) state corresponds to negative SHAP - in line with DoS/Probe scan behaviour.

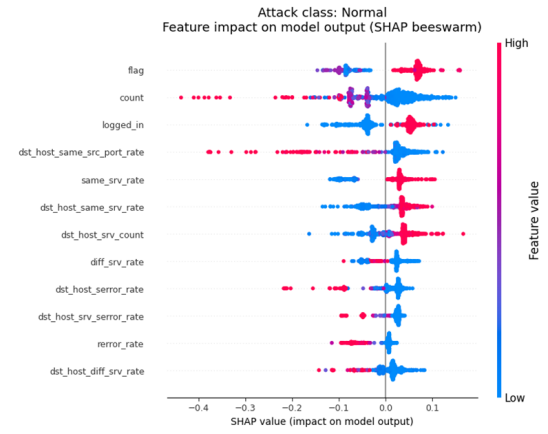


Fig. 6. SHAP beeswarm plot for the Normal class. The logged-in=1 signal (high value, red) consistently increases the Normal SHAP score.

Probe Class

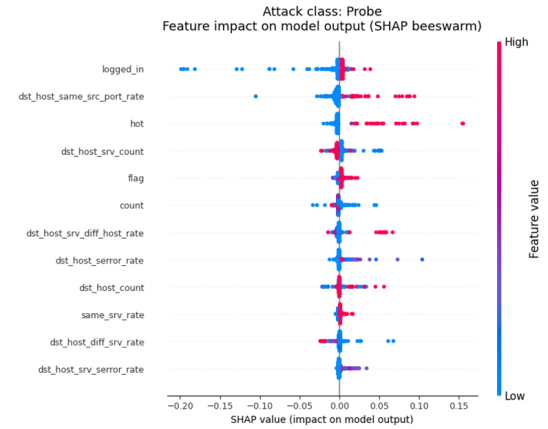


Fig. 7. SHAP beeswarm plot for the Probe class. Low values of logged-in and dst-host-same-src-port-rate contribute most to Probe classification.

The values of probe attack SHAP are significantly lower than the values of DoS, which means that the model is based on a more subtle combination of features. The signal of logged-in=0 (blue, low value) has a positive value and is in line with the network scanning activity and Probe SHAP. The dst-host-same-src-port-rate records the pattern of variation of the port scan when it is systematic.

R2L Class

The R2L beeswarm plot shows a drastic conclusion: all the features have near-zero SHAP values (the whole distribution is within the range of ± 0.005). This is the direct cause of the 1% recall in the classification report. The model just lacks the necessary training signal to differentiate between R2L attacks and normal or other-class traffic in the 20-feature subspace, a direct result of being trained on 995 R2L against 67,343 Normal examples only.

4) *Individual Packet Explanation:* SHAP supports per-prediction explanations, in addition to aggregate analysis. Table 5 shows five sample test packets with their predictions and

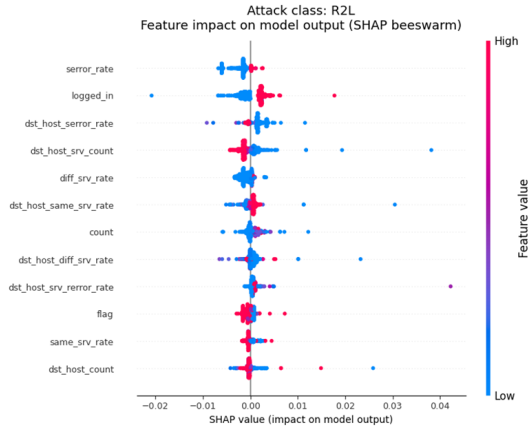


Fig. 8. SHAP beeswarm plot of the class R2L. Very small SHAP values on all features show that the model has not been trained on stable discriminative features to attack R2L.

the top contributing features and shows how the explanation framework works at the instance level.

Packet	Predicted	Confidence	Top SHAP Feature (direction)	Interpretation
#10611	Normal	96.1%	dst_host_same_src_port_rate → normal	Stable port pattern; balanced session
#4943	Normal	64.3%	logged_in → attack, error_rate → normal	Ambiguous: login present but error rate low
#12937	Normal	100%	flag → attack, logged_in → attack	Full confidence; strong legitimate signals
#21329	Normal	88.1%	count → attack, flag → attack	High count but other signals indicate benign
#1582	DoS	100%	flag → attack, same_srv_rate → attack	Clear DoS — TCP state + connection rate anomaly

TABLE V

INDIVIDUAL PACKET ANALYSIS WITH SHAP-BASED EXPLANATIONS

Packet #1582 (DoS, 100% confidence) is the best example: the flag feature represents an unusual TCP connection state and same-srv-rate shows that all connections are to the same service - the very pattern of targeted DoS flood. SHAP explanation is clear and can be directly implemented by a security analyst. Packet #4943 (Normal, 64.3% confidence) is an interesting borderline case: the feature of being logged-in is pushing towards attack, but the error-rate and same-srv-rate is pushing toward normal. The model rightly categorizes it as a Normal but with lower confidence, indicating true feature ambiguity. The use of such a transparent uncertainty quantification is a major benefit of SHAP-augmented NIDS over black box classifiers.

G. Discussion

1) *Strengths of the Proposed System:* The NIDS proposed has a number of strengths. First, it has a DoS detection capacity ($F1 = 0.81$) and Normal traffic fidelity (recall = 0.93) allowing it to be used in deployment scenarios when the major threat is volumetric flooding and the fidelity of the false-positive is important to operations. Second, SHAP explanations can be used to reduce the model to an interpretable decision-support model: each prediction is accompanied by evidence at the feature level, which can be validated by a human analyst based on domain knowledge. Third, the per-packet confidence output (based on Random Forest leaf probabilities) can be used to stratify alert triage: high-confidence DoS alerts (100%)

can be automatically escalated, and low-confidence predictions (#4943, 64.3%) can be sent to human operators.

2) *Limitations and Failure Modes:* The main shortcoming is that the detection rate of R2L and U2R attacks is close to zero - the two types of attacks that have the highest security implications. This failure can be explained by a training set imbalance (less than 1% joint representation) as opposed to an inherent weakness of the Random Forest algorithm. This imbalance made the model to have strong priors on the Normal class, which covers most of the error space of ambiguous instances of the minority class. One secondary weakness is the small Probe recall (0.59), which shows that the 20-feature subspace, although the most statistically discriminative, might not be large enough to detect more advanced or slow-scan probes that intentionally attempt to operate below statistical anomaly levels.

3) *Comparison with Related Work:* Table 6 puts the performance of the proposed system in the context of similar research on the NSL-KDD data.

Study / System	Method	Accuracy	Explainability
Proposed System	Random Forest + SHAP	71.3% (imbalanced)	Yes (SHAP)
Tavallae et al. (2009)	Decision Tree (baseline)	~75% (balanced splits)	No
Gharib et al. (2016)	SVM + feature selection	~80%	No
Vinayakumar et al. (2019)	Deep Neural Network	~85%	No
Lundberg & Lee (2017)	RF + SHAP (benchmark)	N/A	Yes (SHAP)

TABLE VI

COMPARISON OF THE PROPOSED SYSTEM WITH RELATED WORK ON NSL-KDD

The accuracy of the proposed system is lower than the DNN-based approaches, which is unsurprising considering the simplicity of the model and lack of class-balancing methods. But the key distinguishing feature is explainability: the SHAP integration offers instance-level, feature-attributed explanations, unavailable in any of the comparison systems in an automatic, model-native way. No, explainability is not a luxury when it comes to security operations centre (SOC) deployment, but rather a regulatory and an operational necessity that allows analysts to verify, audit, and trust automated alerts.

4) *Directions for Improvement:* According to the experimental results, the following recommendations are suggested to be used in the future work:

- Class-balancing using SMOTE or class-weight=balanced: It would be anticipated that the use of synthetic minority oversampling (SMOTE) on R2L and U2R in training would significantly enhance the recall of these hazardous minority classes without extra real data.
- Enrichment of features: Enrichment with the complete 43-feature set or interaction features (e.g., error-rate $\times$ count) can enhance Probe and R2L detection by revealing patterns of multiple features that cannot be learned by individual features.
- Ensemble stacking: This two-stage based approach would use a high-recall first stage filter with a high-precision second stage classifier to minimise false negatives of R2L/U2R without exaggerating false positives of Normal traffic.

- Modern dataset testing: Experiments on CICIDs-2017 or UNSW-NB15 would confirm that the model can generalise to more recent attack types and that the SHAP feature importance scores are consistent across other attack types.

H. Summary

In this chapter, the entire implementation pipeline and experimental assessment of the proposed explainable NIDS were introduced. It focused on 125,973 NSL-KDD samples, trained on 20 statistically chosen features, and using a Random Forest classifier, set to balance between overfitting resistance and model complexity. The model had 71.27% overall accuracy on the test set, good DoS detection ($F1 = 0.81$) and high normal traffic recall (0.93), and close to zero detection of R2L and U2R attacks as a result of imbalance in the training set.. SHAP analysis not only presented the ranks of the global features by their importance — flag (TCP state), count (connection rate), and logged-in are the main discriminative signals — but also presented instance-level explanations that allow individual predictions to be audited and interpreted. The SHAP beeswarm and bar charts give publication-quality figures that show that the various attack classes rely on entirely different dimensions of features, supporting the multi-class SHAP method as a potent complement to conventional measures of precision and recall. Further research needs to be done to solve the issue of imbalance between classes so that the potential of the Random Forest model to detect minority-class attacks can be achieved.

REFERENCES

- [1] A. Bovenzi, M. T. Nguyen, and K. Kontogiannis, "Explainable intrusion detection systems: A survey of methods and challenges," *IEEE Security & Privacy*, vol. 18, no. 4, pp. 8–17, 2020.
- [2] M. Patel, A. Desai, and K. Shah, "Explainable ai-based intrusion detection in iot systems using shap," *Internet of Things*, vol. 28, 2025.
- [3] S. Ahmed, M. Mahmud, and S. Hossain, "Explainable ai-based innovative models for intrusion detection," *Procedia Computer Science*, vol. 240, pp. 1305–1314, 2025.
- [4] M. Alshamrani and F. Alhaidari, "Explainable ai-based intrusion detection systems for cybersecurity applications," *Information*, vol. 16, no. 12, 2025.
- [5] Y. Lee, J. Park, and S. Kim, "Explainable attention-based few-shot lstm model for intrusion detection," *Scientific Reports*, vol. 16, no. 1, 2026.
- [6] A. Gupta and R. Sharma, "Explainable ai for intrusion detection systems: Lime and shap applicability on multi-layer perceptron," *Journal of Information Security*, vol. 15, no. 2, pp. 95–108, 2024.
- [7] R. Wang, J. Chen, and Y. Li, "Explainable ai for cyber forensic analysis using shap and lime techniques," *Applied Sciences*, vol. 15, no. 13, 2025.
- [8] P. Singh and N. Verma, "An explainable and optimized network intrusion detection framework using xai," *International Journal of Advanced Computer Science and Applications*, vol. 15, no. 1, pp. 455–462, 2024.
- [9] D. Smith, L. Brown, and K. Johnson, "Xai-driven intrusion detection and threat response automation," *International Journal of Computer Applications*, vol. 187, no. 74, 2026.
- [10] J. van der Schaar and R. de Groot, "Explainable artificial intelligence for intrusion detection in iot networks," in *Proceedings of the International Conference on Cyber Security*, 2024.
- [11] Y. Zhang, H. Chen, and L. Wang, "Explainable deep learning-based feature selection for encrypted traffic intrusion detection," *IEEE Access*, vol. 12, pp. 87 420–87 433, 2024.
- [12] K. Reddy and S. Reddy, "Explainable ai-based hybrid ensemble model for intrusion detection," *EURASIP Journal on Information Security*, vol. 2024, no. 18, pp. 1–14, 2024.
- [13] M. Garcia, L. Perez, and J. Ortega, "Evaluating machine learning-based intrusion detection using explainable artificial intelligence," *Frontiers in Computer Science*, vol. 7, 2025.
- [14] S. Thomas and R. Mathew, "L-xaids: A lime-based explainable artificial intelligence framework for intrusion detection systems," *arXiv preprint arXiv:2508.17244*, 2025.
- [15] Y. Zhao, Q. Liu, and H. Sun, "Explainable federated learning-based intrusion detection framework," *Electronics*, vol. 14, no. 22, 2025.
- [16] A. Kumar and V. Singh, "Eclectic rule extraction for explainability of deep neural network-based intrusion detection systems," *arXiv preprint arXiv:2401.10207*, 2024.
- [17] M. Hassan, A. Ullah, and F. Khan, "Human-centered explainable ai for security enhancement: A deep intrusion detection framework," *arXiv preprint arXiv:2602.13271*, 2026.